

BRACU Machine Learning Lecture Notes

Topic: Sequence-to-Sequence, Attention & Transformers

1. Sequence-to-Sequence Models:

Sequence-to-Sequence models map an input sequence to an output sequence using an encoder and a decoder.

2. Attention Mechanism:

The Attention Mechanism solves the bottleneck by allowing the decoder to dynamically attend to all encoder states.

3. Scaled Dot-Product Attention:

The Attention formula is $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$. Scaling by $\sqrt{d_k}$ prevents the softmax from being too peaked.

4. BERT Pretraining Objectives:

BERT is an encoder-only model pre-trained on Masked Language Modeling (MLM, predicting 15% masked tokens).

5. Variational Autoencoders (VAEs):

VAEs use the Reparameterization Trick $z = \mu + \sigma * \epsilon$ to make stochastic sampling differentiable.

6. Generative Adversarial Networks (GANs):

GANs train a Generator G and a Discriminator D in a minimax zero-sum game: $\min_G \max_D V(D, G)$.